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 MitchellGray100

- Developed **Python**, **Rust**, and **Bash** code on various NDA projects · Updated and created **CI/CD** pipelines · Created GitLab **REST API** live data visualizations for clients · Used ArgoCD and Helm to deploy AWS EKS cluster.
- Built internal analytics tools using **Python**, **Neo4j**, NeoDash, and PageRank-based graph analysis to generate engineering metrics and documentation adopted across multiple teams within the company.

Relevant Academic Work Experience

Georgia Tech Contextual Computing Group

Atlanta, GA

Graduate Researcher | AI, ML, Python, Typescript

June 2025 – Aug 2026

- Worked on Nosi, an IDE that combines an encrypted virtual file system and application behavior-logging · Utilizes **Machine Learning** techniques to analyze behavioral data and flag potentially-cheating students.
- Published at **Learning at Scale 2026** · Integrated the IDE into multiple Georgia Tech graduate-level CS courses.

Georgia Tech

Atlanta, GA

Graduate Teaching Assistant | AI, Python, NumPy

May 2025 – Aug 2026

- Taught graduate-level **Artificial Intelligence** · Helped with new plagiarism-detection tooling integration.

Cornell University

Ithaca, NY

Undergraduate Teaching Assistant | Python, NumPy, SQL, Java

Aug 2022 – May 2024

- Received the **Course Staff Exceptional Service Award** · Taught Databases, Object-Oriented Programming, and Physics · Lectured 400+ students with custom-designed lecture content · Led Exam sessions and other TAs.

Cornell Database Group

Ithaca, NY

Undergraduate Researcher | ML, Java, AWS

Jan 2022 – May 2024

- Published and presented paper at **VLDB 2023** combining **Reinforcement Learning** and Worst-Case Optimal Joins.
- Worked on upgrading query engine to a **Distributed Architecture** · Created dynamic statistics & RL visualizations.

Publications

Learning at Scale (2026)

[<https://dl.acm.org/doi/abs/10.14778/3611540.3611629>]

Nosi IDE: Securing Coding Assessment Integrity in the Age of LLMs via Process-Driven Analytics

VLDB (2023)

[<https://dl.acm.org/doi/abs/10.14778/3611540.3611629>]

Demonstrating ADOPT: Adaptively Optimizing Attribute Orders for Worst-Case Optimal Joins via Reinforcement Learning

Awards

Oracle, Database Organization

Mentorship and Leadership Excellence Award

Recognizes outstanding efforts in mentorship and training of others that go above and beyond the call of duty. Multi-day, multi-recipient efforts beyond the direct organization.

Cornell Bowers CIS

Course Staff Exceptional Service Award

Faculty-nominated award for exceptional commitment, hard-work, dedication, and exceptional service of a Teaching Assistant during the 2023-2024 academic year.

Relevant Projects

New AI Infra Project Placeholder | Personal Project | AI, Go | [Github](#)

May 2026 – Present

- 1 of 44 projects selected for BOOM 2024. My team and I created a Scalable Platform for Efficient Execution of Distributed testing. The fault-tolerant system contains a controller node that orchestrates worker nodes that run **JUnit** tests on **Java** code. The worker nodes report their findings to the controller. Once all tests are ran, test results are shown in the frontend to the user.

Improving 3D Point-Cloud Classification | Deep Learning | AI, Python, Pytorch | [Github](#) Aug 2025 – Dec 2025

- For this project, we re-implemented and upgraded state-of-the-art point cloud classification Deep Neural Networks. We started by implementing PointNet and PointPillars. Then we upgraded them by modifying their architecture, adding Dropout, implementing Data augmentation, and introducing Residual Connections. We improved validation accuracy by 4% and massively reduced overfitting.

Distributed Testing Platform (SPEED) | Masters Project | Java | [Github](#) | [BOOM](#)

Aug 2023 – May 2024

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SpGEMM Parallelization on GPUs | Parallel Computing | CUDA, C++ | [Github](#)

Jan 2024 – May 2024

- In this project, our team parallelized sparse-matrix multiplication operations using GPUs and experimented with multiple variations of the algorithm, massively improving performance of the algorithms.